

DSA210 Final Project Report

Purchasing-Power-Adjusted Developer Salaries

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Short summary

In this project I looked at developer salaries from the Stack Overflow Developer Survey 2024. I did not only use raw salary, because raw salary can be misleading when countries have very different living costs. To make the comparison more realistic, I joined the survey data with a country-level cost-of-living dataset and created a purchasing-power-adjusted salary variable.

After preparing the data, I did exploratory data analysis, hypothesis tests, and a small machine learning part. The ML task was to predict whether a developer belongs to the high adjusted salary group or not. I tried to keep the analysis connected to the methods we covered in class, instead of adding advanced models just for the sake of adding them.

1. Motivation

I picked this topic because developer salary is a common topic, but I think raw salary by itself is not always fair to compare. A salary that looks high in one country may not feel as high after rent, food, and daily expenses. In another country, a lower salary may actually give more real purchasing power.

Because of that, the main idea of my project is simple: compare developer earnings after adjusting for country-level cost of living. This also fits the course guideline, because I used a public dataset and enriched it with another public dataset.

The main questions I tried to answer were these:

- How does the salary picture change after using a cost-of-living adjustment?
- Do adjusted salaries differ between work settings, experience groups, and education groups?
- Is job satisfaction related to adjusted salary?
- Can basic machine learning models predict high adjusted salary better than a simple baseline?

2. Data source and collection

The main dataset is the Stack Overflow Developer Survey 2024. I used the public survey file, especially the columns about country, compensation, work setting, education, experience, job satisfaction, employment, and developer role.

The second dataset is `Cost_of_Living_Index_by_Country_2024.csv`. I used this file to add country-level cost-of-living information. The important point is that the Stack Overflow data is individual-level survey data, while the cost-of-living file is country-level data. I matched them by country name.

The match worked better than I expected. After filtering to professional developers with positive salary, there were 21,549 rows. Out of these, 21,405 rows matched with the cost-of-living data. That is a 99.33 percent match rate.

Item	Value
Original Stack Overflow rows	65,437
Professional developers with positive salary	21,549
Rows matched with cost-of-living data	21,405
Rows not matched	144
Country match rate	99.33%
Rows with job satisfaction value	15,003

3. Data analysis

3.1 Cleaning and preparation

The raw data was not ready to analyze directly. First, I selected the variables I needed. Then I kept professional developers with positive yearly compensation, because salary is the main outcome in this project. Rows without salary could still be interesting in general, but they cannot be used for adjusted salary analysis.

Then I matched the country names between the survey data and the cost-of-living file. After that, I created the adjusted salary variable. The idea was to divide the raw salary by the cost-of-living index and multiply by 100. So if a country has a higher cost-of-living index, the same salary becomes less strong after adjustment.

I also created some grouped variables. For example, I grouped years of professional coding experience and education levels so that the plots and hypothesis tests would be easier to read. For machine learning, I created a high or low adjusted salary target based on the median adjusted salary.

For the ML models, I used basic preprocessing. Missing numeric values were filled with the median. Missing categorical values were filled with the most common value. Categorical variables were one-hot encoded. I scaled the numerical variables for kNN and logistic regression, because these models are sensitive to feature ranges.

3.2 Exploratory data analysis

I started the EDA by checking the size of the data, missing values, salary distributions, and group summaries. Salary data was clearly skewed. This is why I looked at medians together with means. In this kind of data, the mean can be pulled up by a small number of very high salaries.

The median raw yearly salary was 66,066. After cost-of-living adjustment, the median adjusted salary was 169,038.22. The adjusted value is not meant to be a perfect real-life purchasing power number. I use it as a simple way to compare countries more fairly than raw salary alone.

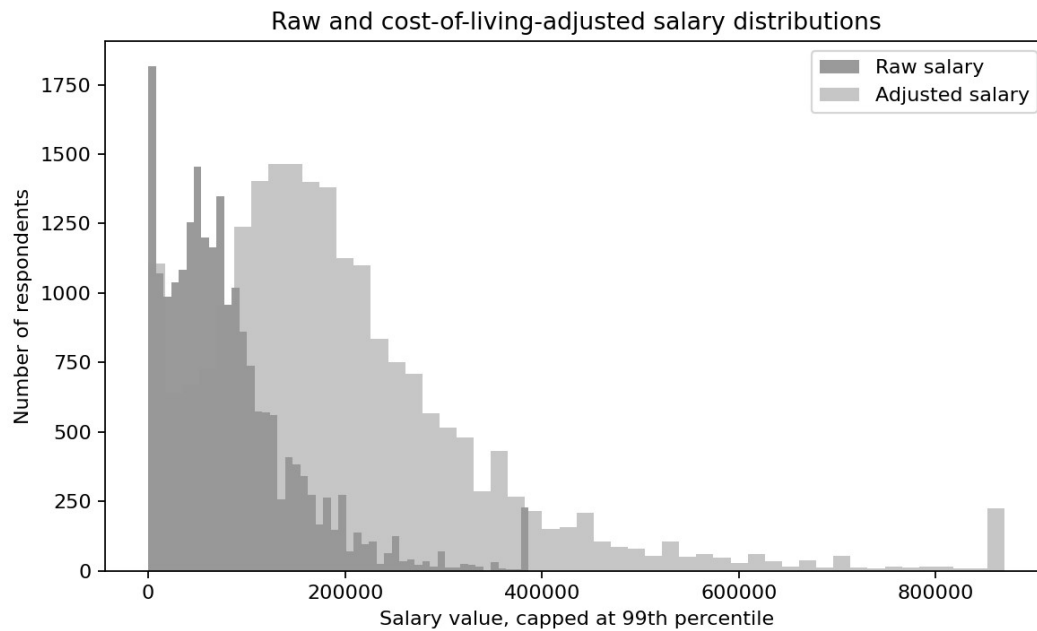


Figure 1. Raw salary and adjusted salary distributions.

The distribution plot shows why salary should be treated carefully. Both raw and adjusted salaries have long right tails. So I avoided relying only on average salary.

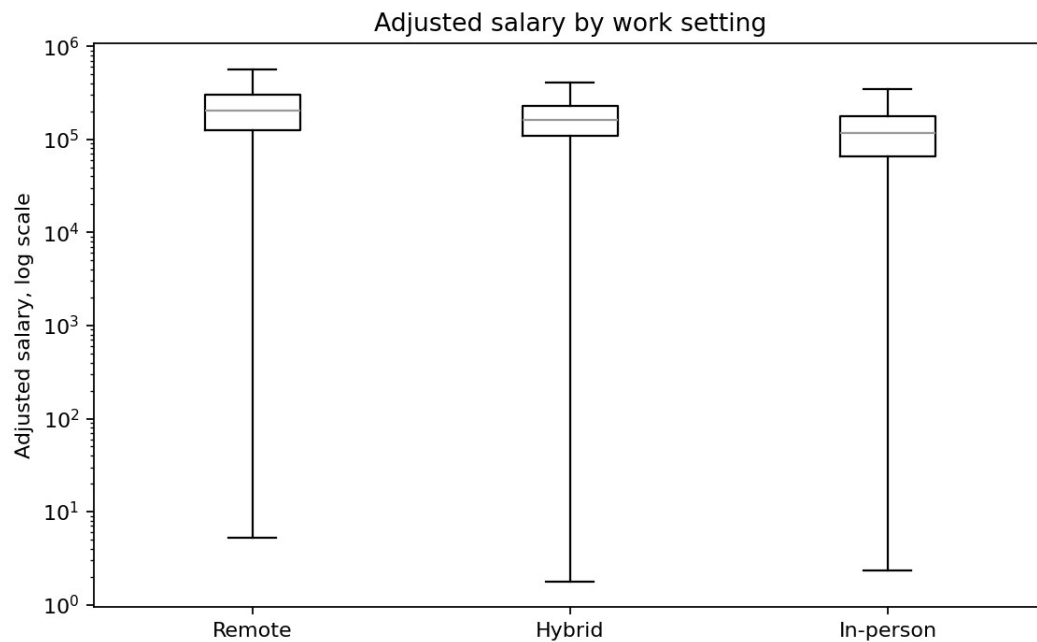


Figure 2. Adjusted salary by work setting.

Remote developers had the highest median adjusted salary in this dataset. Hybrid developers were in the middle, and in-person developers had the lowest median. I am careful with this result, because it does not mean remote work itself causes higher salary. It only shows a relationship in the survey data.

Work setting	Count	Median adjusted salary
Hybrid (some remote, some in-person)	8,947	162,174
In-person	3,429	116,299
Remote	9,021	203,180

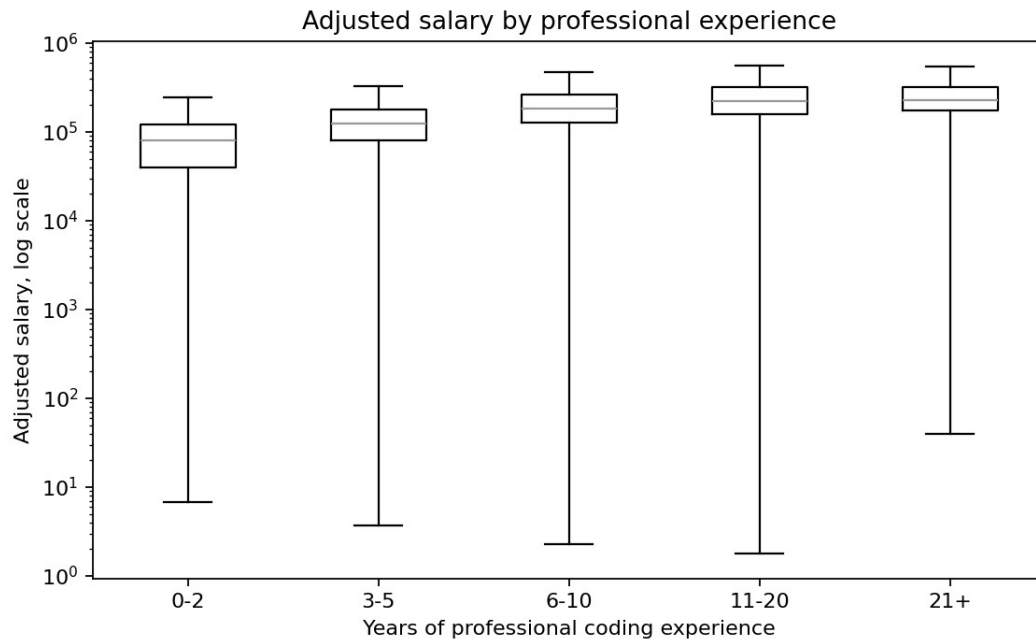


Figure 3. Adjusted salary by experience group.

Experience had one of the clearest patterns. More experienced developers generally had higher adjusted salaries. This also made sense in the ML part, where professional coding experience became the most important random forest feature.

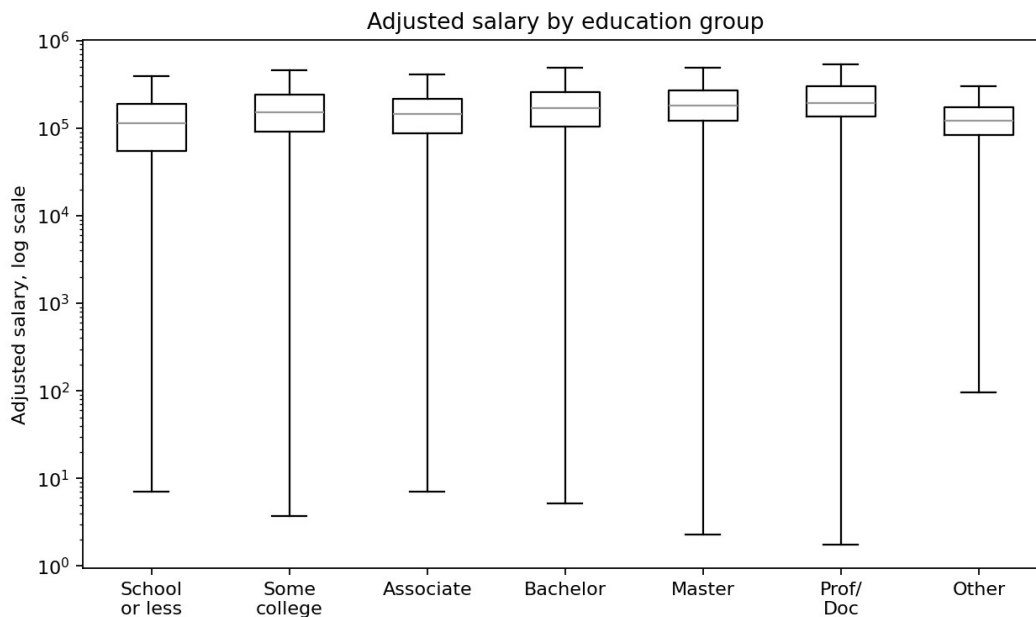


Figure 4. Adjusted salary by education group.

Education groups also had differences, but the difference did not look as strong as the experience difference. This is why I did not want to overstate education as the main driver.

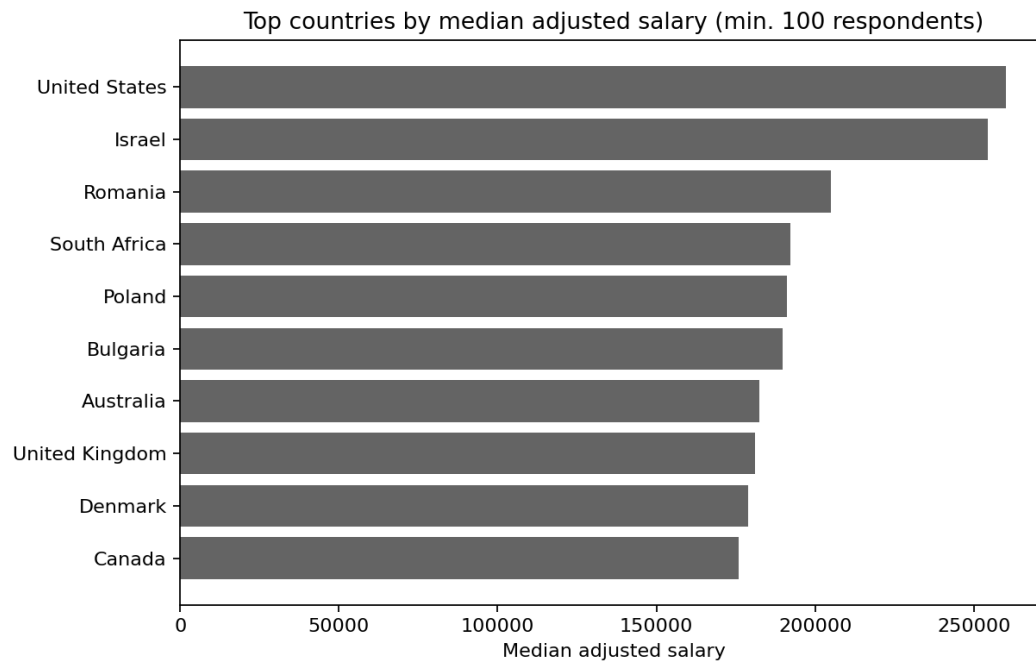


Figure 5. Top countries by median adjusted salary after filtering for enough observations.

Country	Count	Median adjusted salary
United States	4,256	260,159
Israel	213	254,456
Romania	159	205,074
South Africa	155	192,336
Poland	545	191,261
Bulgaria	101	189,695
Australia	455	182,469
United Kingdom	1,274	181,210

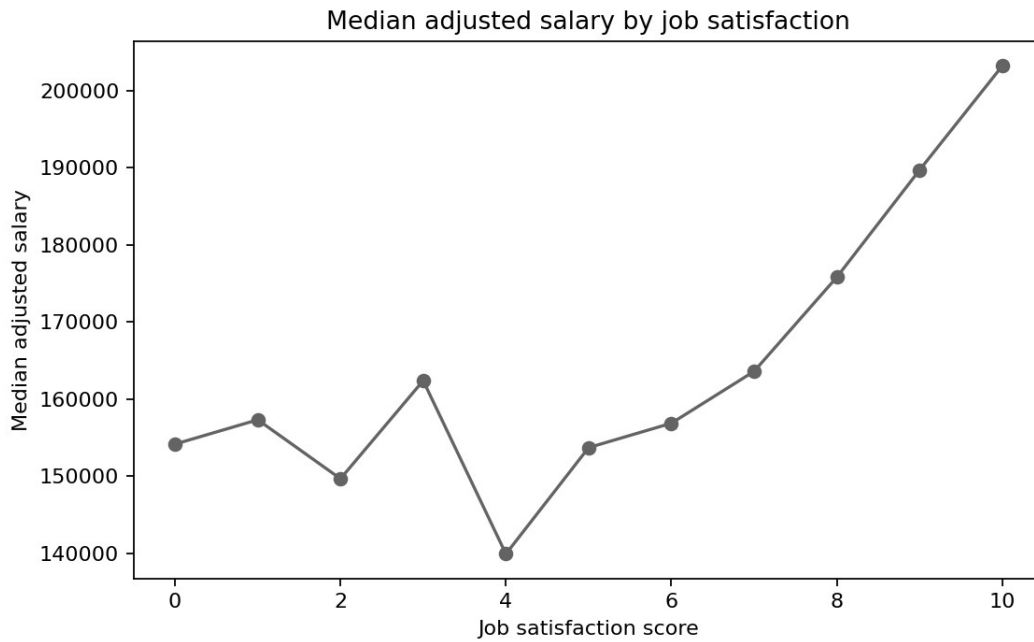


Figure 6. Job satisfaction and adjusted salary.

Job satisfaction had a positive relationship with adjusted salary, but it was weak. So salary seems to matter a little, but it is clearly not the only thing related to job satisfaction.

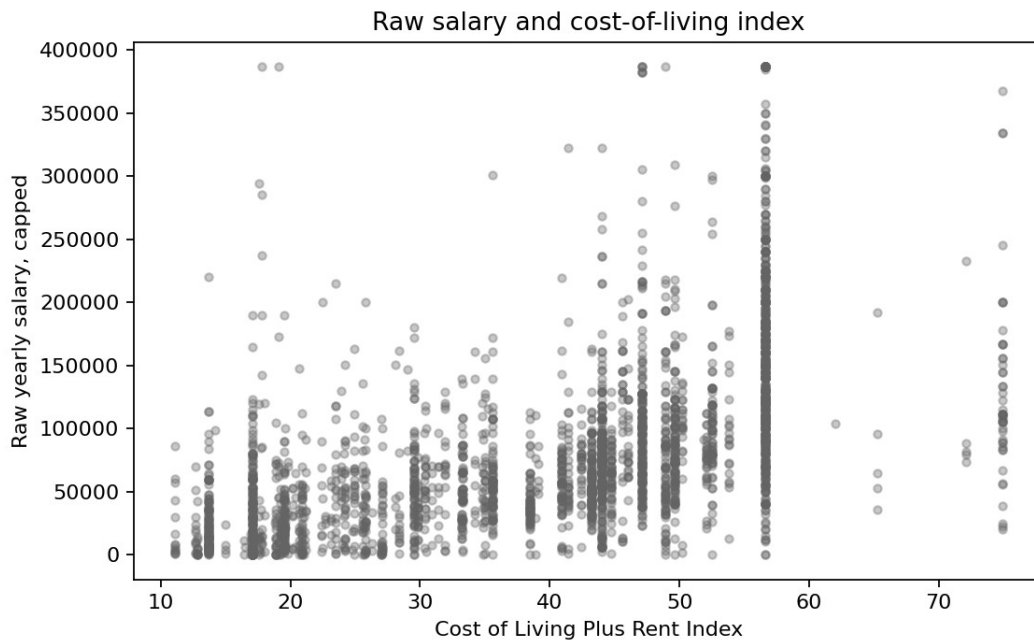


Figure 7. Cost-of-living index and raw salary.

3.3 Hypothesis tests

For the hypothesis tests, I mostly used non-parametric tests because the adjusted salary distribution was skewed and had outliers. This felt safer than assuming normal salary distributions. I used $\alpha = 0.05$. I also used the wording from the course: reject H_0 or fail to reject H_0 .

The tests I used were Kruskal-Wallis tests for group differences, Spearman correlation for job satisfaction, and a chi-square test for the relationship between work setting and high or low adjusted salary.

Test	Question	Statistic	p-value	Decision
Kruskal-Wallis	Salary differs across work settings	1305.12	< 0.001	reject H0
Kruskal-Wallis	Salary differs across experience groups	5505.08	< 0.001	reject H0
Kruskal-Wallis	Salary differs across education groups	580.84	< 0.001	reject H0
Spearman correlation	Job satisfaction and salary are related	0.1259	< 0.001	reject H0
Chi-square independence	Work setting and high/low salary are associated	1176.28	< 0.001	reject H0

All listed tests rejected H0 at $\alpha = 0.05$. The strongest practical pattern was the experience group difference. Work setting and education also had statistically significant differences. The Spearman result for job satisfaction was significant too, but the rho value was small, so I interpreted it as a weak relationship.

3.4 Machine learning

For the machine learning part, I used a binary classification task. The target was high adjusted salary versus low adjusted salary. I defined high adjusted salary using the median adjusted salary in the ML dataset.

To keep the run practical and reproducible, the ML part used 5,000 cleaned rows. The training set had 3,750 rows and the test set had 1,250 rows. I used a stratified split so that the high and low classes stayed balanced.

I compared five models: a baseline model, logistic regression, kNN, decision tree, and random forest. I chose these because they match the course topics. Logistic regression is a simple binary classifier. kNN uses nearest neighbors, so scaling is important. Decision trees are easy to interpret but can overfit. Random forest combines many trees and usually gives better performance than one tree.

I did not rely only on accuracy. I used accuracy, precision, recall, F1-score, and confusion matrices. I mainly looked at F1-score because it balances precision and recall.

Model	Accuracy	Precision	Recall	F1
Random Forest	0.756	0.734	0.799	0.765
Logistic Regression	0.758	0.755	0.760	0.758
Decision Tree	0.738	0.708	0.804	0.753
kNN	0.726	0.709	0.763	0.735
Baseline	0.503	0.000	0.000	0.000

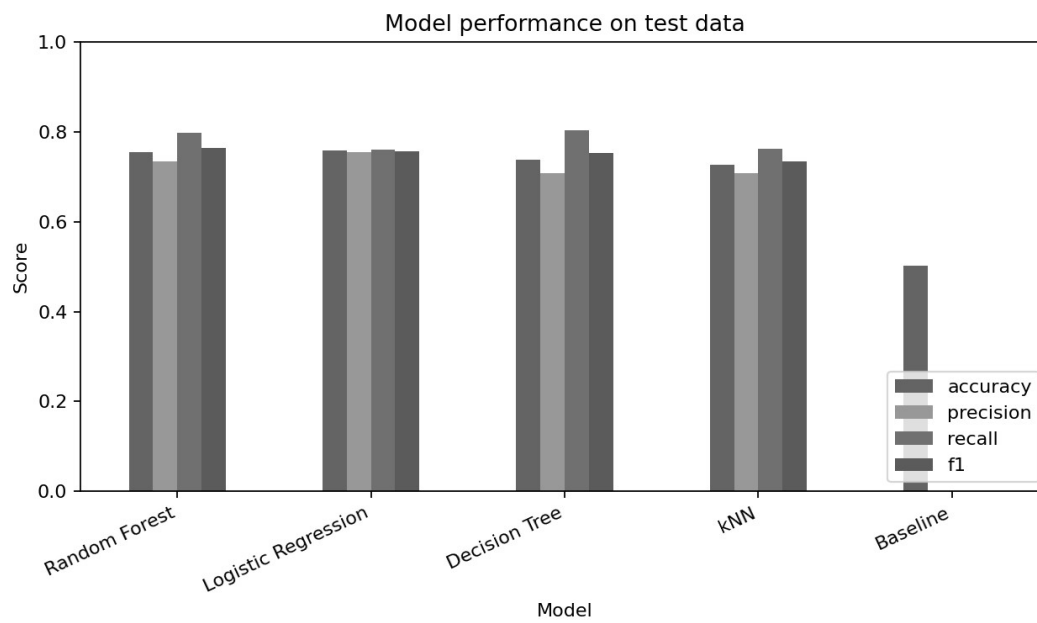


Figure 8. Model metric comparison on the test set.

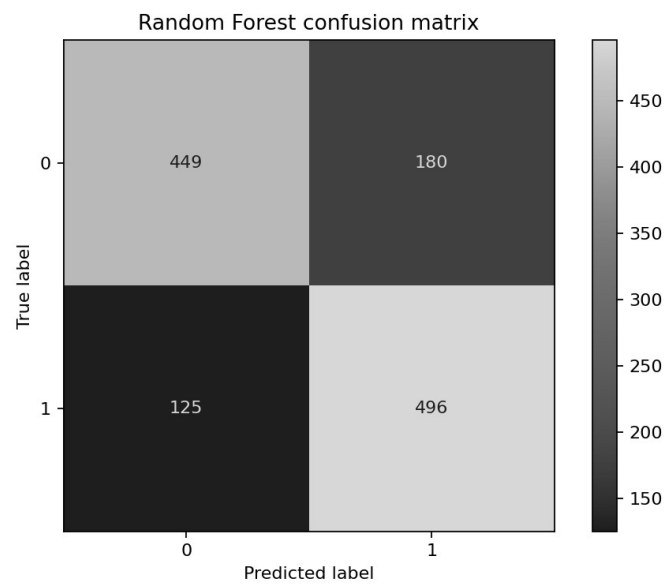


Figure 9. Random forest confusion matrix.

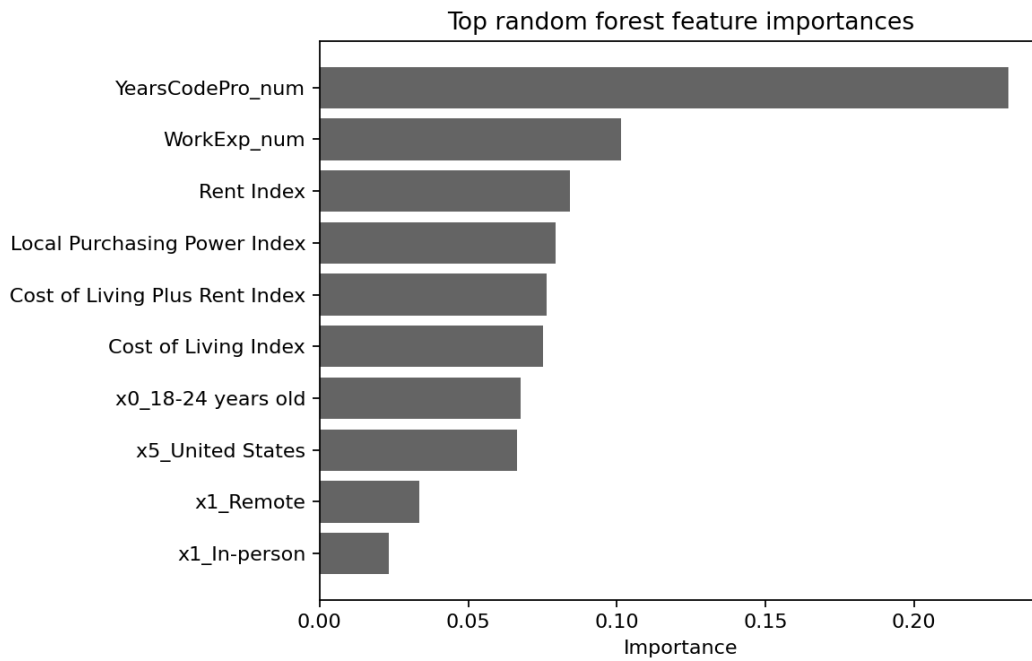


Figure 10. Random forest feature importance.

Random Forest had the best F1-score, 0.765. Logistic Regression was very close with 0.758. I found this interesting because the more complex model was best, but a simpler model still performed well. This means the classification problem has some clear signal, but it is not a perfect prediction problem.

Feature	Importance
YearsCodePro_num	0.232
WorkExp_num	0.101
Rent Index	0.084
Local Purchasing Power Index	0.079
Cost of Living Plus Rent Index	0.076
Cost of Living Index	0.075
x0_18-24 years old	0.068
x5_United States	0.066

The feature importance result also matched the EDA. Professional coding experience was the most important feature. Work experience and country-level cost variables were also important. This makes sense because the target is about adjusted salary, not only raw salary.

4. Findings

- Cost-of-living adjustment changed the way salaries should be read. Raw salary alone was not enough for comparing developers across countries.
- Remote developers had the highest median adjusted salary in this dataset, but this should be read as association only.
- Experience was the clearest group pattern. Developers with more professional experience generally had higher adjusted salary.

- Education groups had statistically significant differences, but the practical difference looked smaller than the experience difference.
- Job satisfaction and adjusted salary had a positive but weak relationship.
- The ML models performed clearly better than the baseline. Random Forest had the best F1-score, while Logistic Regression was close and easier to explain.

5. Limitations and future work

There are several limitations in this project. The Stack Overflow survey is public and useful, but it is still a survey. It is not a random sample of every developer in the world. Developers who answer the survey may be different from developers who do not answer it.

- Salary is self-reported, so it can include mistakes or different interpretations of compensation.
- Many people do not report salary, and this missingness may not be random.
- The cost-of-living dataset is country-level. It does not capture city-level differences inside the same country.
- The adjusted salary formula is simple. It is better than raw salary alone, but it is not a perfect measure of real purchasing power.
- The analysis is observational. I should not claim that remote work, education, or experience causes higher salary.
- The ML part used a 5,000-row sample for runtime and reproducibility. A future version could train on the full cleaned dataset.
- Time-series models were not used because the data is cross-sectional, not repeated observations over time.
- Causal methods like IPW or propensity score matching were not used because this project does not have a clean treatment variable.

For future work, I would try to add city-level cost-of-living data. That would probably improve the adjustment a lot. I would also like to use several years of Stack Overflow survey data. With multiple years, it would become more reasonable to analyze trends over time. Another possible extension is a separate job satisfaction project, because salary explained only a small part of it here.

6. Conclusion

Overall, this project gave a more realistic view of developer salaries by combining survey data with cost-of-living data. The main result is that raw salary and adjusted salary are not the same story. Cost of living changes how we should compare countries and developer groups.

The EDA, hypothesis tests, and ML models all pointed in a similar direction. Experience, work setting, and country-level economic context were important for adjusted salary. The random forest model performed best, but logistic regression was also strong, which made the results easier to trust and explain.

I see the project as a full data science workflow: collect and enrich data, clean it, visualize it, test group differences, build prediction models, and then explain the limits of the results. The most important caution is that these are associations and predictions, not causal effects.

References and data sources

- Stack Overflow Developer Survey 2024, public survey dataset. I used `survey_results_public.csv` and `survey_results_schema.csv`.
- `Cost_of_Living_Index_by_Country_2024.csv`, public country-level cost-of-living dataset.
- DSA210 lecture and recitation materials. I used the course methods for EDA, visualization, hypothesis testing, model evaluation, ML, causal interpretation, and time-series method choice.